

Process Control And

To impart knowledge on measurement of various parameters and instrumentation for

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5123C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5123C.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Wood and Paper Technology
Course code	5123C
Course title	Process Control And
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To impart knowledge on measurement of various parameters and instrumentation for processes in wood based industries.
- To provide knowledge of process control.

Course outcomes

CO	Official outcome	Study level
CO1	static characteristics of instruments in wood based 15 Understanding industries Describe the working principle of measuring	See official syllabus
CO2	14 Understanding instruments in process plants Outline the measuring instruments in paper	See official syllabus
CO3	15 Understanding industry Explain the modes of control action and the	See official syllabus
CO4	14 Understanding computer control system	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	instruments in wood based industries.	4	As specified
Module 2	plants	4	As specified
Module 3	Outline the measuring instruments in paper industry	3	As specified
Module 4	Explain the modes of control action and the computer control system	4	As specified

MODULE 1

instruments in wood based industries.

This module is presented from the official syllabus scope for Process Control And. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: instruments in wood based industries.
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

1 Understanding industries

1 Understanding industries is an official topic in Module 1 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding industries using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding industries should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding industries means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 1 Understanding industries
 definition/meaning .
 application
 Technical terms English-
 .

Construct the instrumentation block diagram

Construct the instrumentation block diagram is an official topic in Module 1 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Construct the instrumentation block diagram using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, construct the instrumentation block diagram should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Construct the instrumentation block diagram means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓൺലൈൻ കോഴ്സ് നിർമ്മിക്കുന്നതിനുള്ള സോഫ്റ്റ്‌വെയർ ഉപയോഗിക്കുന്നതിനുള്ള നിർദ്ദേശങ്ങൾ. നിർദ്ദേശങ്ങൾ നിർമ്മിക്കുന്നതിനുള്ള, steps, application നിർദ്ദേശങ്ങൾ നിർമ്മിക്കുന്നതിനുള്ള. Technical terms English-ൽ നിർമ്മിക്കുന്നതിനുള്ള.

Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic

Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic is an official topic in Module 1 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, suggest the functions of measuring instruments 6 understanding analyze different static and dynamic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-
6 Understanding Analyze different static and dynamic
definition/meaning . . . , steps, application
 Technical terms English-
 .

7 Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the working principle of measuring instruments in process

7 Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the working principle of measuring instruments in process is an official topic in Module 1 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the

working principle of measuring instruments in process using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 understanding characteristics of instruments importance of instrumentation in process industries; block diagram; functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. transducers, their classification and selection criteria; on and offline measurements. static characteristics- accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone dynamic characteristics, speed of response, fidelity, lag, dynamic error. describe the working principle of measuring instruments in process should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the working principle of measuring instruments in process means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 7 Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of

measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the working principle of measuring instruments in process industries. definition/meaning. steps, application. Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

instruments in wood based industries.		
	Explain the importance of Instrumentation in	
M1.01		1
Understanding	industries	
	Construct the instrumentation block diagram	
M1.02		1
Understanding		
	Suggest the functions of measuring instruments	
M1.03		6
Understanding	Analyze different static and dynamic	
M1.04		7
Understanding	characteristics of instruments	
Contents:		
Importance of instrumentation in process industries;		
Block diagram; Functions of measuring instruments, signaling, transmitting, registering,		
indicating and recording - elements of measuring instruments, primary, secondary,		
manipulating and functioning element.		
Transducers, their classification and selection criteria; On and offline measurements.		
Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold,		
hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

plants

This module is presented from the official syllabus scope for Process Control And. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: plants
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding measuring instruments Discuss the working principle of Pressure

3 Understanding measuring instruments Discuss the working principle of Pressure is an official topic in Module 2 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding measuring instruments Discuss the working principle of Pressure using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding measuring instruments discuss the working principle of pressure should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding measuring instruments Discuss the working principle of Pressure means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding measuring instruments

Discuss the working principle of Pressure പ്രഷർ definition/meaning പ്രഷർ ന്റെ അർത്ഥം . $\text{പ്രഷർ ന്റെ പ്രവർത്തന തത്വം}$, steps, application $\text{പ്രഷർ ന്റെ പ്രയോഗം}$ പ്രഷർ ന്റെ ഘടന . Technical terms English- പ്രഷർ പ്രഷർ മീറ്റർ .

3 Understanding measuring instruments Outline the working principle of flow and level

3 Understanding measuring instruments Outline the working principle of flow and level is an official topic in Module 2 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding measuring instruments Outline the working principle of flow and level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding measuring instruments outline the working principle of flow and level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding measuring instruments Outline the working principle of flow and level means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding measuring instruments

Outline the working principle of flow and level മെഴുവിലിന്റെ പ്രവർത്തനം

definition/meaning മെഴുവിലിന്റെ അർത്ഥം. മെഴുവിലിന്റെ പ്രവർത്തനം, steps, application

മെഴുവിലിന്റെ പ്രവർത്തനം മെഴുവിലിന്റെ. Technical terms English-മെഴുവിലിന്റെ പ്രവർത്തനം.

3 Understanding measuring instruments Illustrate the working principle of PH, viscosity

3 Understanding measuring instruments Illustrate the working principle of PH, viscosity is an official topic in Module 2 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding measuring instruments Illustrate the working principle of PH, viscosity using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding measuring instruments illustrate the working principle of ph, viscosity should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding measuring instruments Illustrate the working principle of PH, viscosity means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Understanding measuring instruments

Illustrate the working principle of PH, viscosity
 definition/meaning
 .
 , steps, application
 .
 Technical terms English-
 .

5 Understanding and humidity measuring instruments Temperature measuring instruments - Principle and working of -Thermometers, filled system and vapor pressure thermometers -Thermocouples, types, materials - resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer.

5 Understanding and humidity measuring instruments Temperature measuring instruments - Principle and working of -Thermometers, filled system and vapor pressure thermometers -Thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer. is an official topic in Module 2 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding and humidity measuring instruments
Temperature measuring instruments - Principle and working of -Thermometers, filled system and vapor pressure thermometers -Thermocouples, types, materials - resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter – Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding and humidity measuring instruments temperature measuring instruments - principle and working of -thermometers, filled system and vapor pressure thermometers -thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. pressure measuring instruments - principle and working of - bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. flow measuring devices: coriolis mass flow meter, ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. level measuring devices: float level indicator, magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. ph measurement, ph scale, reference and measuring electrode - conductivity measurement, electrical conductivity meter – density measurement, online density meter, hydrometer. humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type viscosity - absolute and kinematic viscosity - viscosity

measurement - say bolt and redwood viscometer - falling- ball viscometer. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding and humidity measuring instruments Temperature measuring instruments - Principle and working of -Thermometers, filled system and vapor pressure thermometers - Thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer.Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 5 Understanding and humidity measuring instruments Temperature measuring instruments - Principle and working of - Thermometers, filled system and vapor pressure thermometers -Thermocouples, types, materials -resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer.Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt

and Redwood viscometer - Falling- ball viscometer.
 definition/meaning, steps, application
 Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

plants		
	Explain the working principle of Temperature	
M2.01		3
Understanding	measuring instruments	
	Discuss the working principle of Pressure	
M2.02		3
Understanding	measuring instruments	
	Outline the working principle of flow and level	
M2.03		3
Understanding	measuring instruments	
	Illustrate the working principle of PH, viscosity	
M2.04		5
Understanding	and humidity measuring instruments	
	Series Test - I	1
Contents:		
Temperature measuring instruments - Principle and working of -Thermometers, filled		
system and vapor pressure thermometers -Thermocouples, types, materials - resistance		
thermometers, thermistor - pyrometer, radiation and optical.		
Pressure measuring instruments - Principle and working of - Bourdon tube, bellows,		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Outline the measuring instruments in paper industry

This module is presented from the official syllabus scope for Process Control And. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the measuring instruments in paper industry
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

5 Understanding paper industries

5 Understanding paper industries is an official topic in Module 3 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding paper industries using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding paper industries should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding paper industries means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-5 Understanding paper industries

പാപ്പർ ഇൻഡസ്ട്രി ഓരോന്നിന്റെയും definition/meaning എഴുതുക. പാപ്പർ ഇൻഡസ്ട്രി ഉണ്ടാക്കുന്ന steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement

Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement is an official topic in Module 3 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss the properties of paper 5 understanding illustrate the working principle for measurement should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Discuss the properties of paper 5

Understanding Illustrate the working principle for measurement

definition/meaning . steps, application . Technical terms English- .

5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper

5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper is an official topic in Module 3 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding of properties of paper definitions and methods of determination of : physical properties - grammage, caliper, bulk, smoothness and porosity; mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: optical properties: brightness, opacity, gloss and color: resistance properties: water absorptiveness, cobb test, oil absorbency, air/gas permeability. electrical properties: dielectric constant, dielectric strength, dielectric loss and properties of electrical insulating paper should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Outline the measuring instruments in paper industry		
Suggest various process parameters in wood and		
M3.01		5
Understanding		
	paper industries	
M3.02	Discuss the properties of paper	5
Understanding		
	Illustrate the working principle for measurement	
M3.03		5
Understanding		
	of properties of paper	

Contents:

Definitions and methods of determination of :

Physical properties - Grammage, caliper, bulk, smoothness and porosity;

Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending

stiffness:

Optical Properties: brightness, opacity, gloss and color:

Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability.

Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Explain the modes of control action and the computer control system

This module is presented from the official syllabus scope for Process Control And. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the modes of control action and the computer control system
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding control.

3 Understanding control. is an official topic in Module 4 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding control. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding control. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding control. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Understanding control. ഏകീകൃതമായ നിയന്ത്രണ
definition/meaning നിയന്ത്രണത്തിന്റെ അർത്ഥം. നിയന്ത്രണത്തിന്റെ ഘട്ടങ്ങൾ, steps, application
നിയന്ത്രണത്തിന്റെ പ്രയോഗം. Technical terms English-നിയന്ത്രണത്തിന്റെ പദങ്ങൾ.

Comprehend computerized control system.

Comprehend computerized control system. is an official topic in Module 4 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Comprehend computerized control system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, comprehend computerized control system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Comprehend computerized control system. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓം Comprehend computerized control system.

ഓംഓംഓംഓംഓം ഓംഓം definition/meaning ഓംഓംഓംഓംഓംഓം. ഓംഓംഓം ഓംഓംഓം ഓംഓംഓം,
steps, application ഓംഓംഓം ഓംഓംഓംഓം ഓംഓംഓം. Technical terms English-ഓം ഓംഓംഓം
ഓംഓംഓംഓംഓം.

Summarize final control elements.

Summarize final control elements. is an official topic in Module 4 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize final control elements. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize final control elements. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize final control elements. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
Summarize final control elements.

definition/meaning .
steps, application . Technical terms English-
.

Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed- back, controllers - proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., "Introduction to Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard", Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., "Pulp and Paper Industries Technology and R6 Instrumentation", Kotharis Desk book series, 1995. G. Stephanopoulous, 'Chemical Process Control - Theory and Practice', R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: SI.No Website Link 1 <https://www.nptel.ac.in/courses/103103037/> [https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2 notes-ics-pdf-notes/](https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2-notes-ics-pdf-notes/)

Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed- back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., "Introduction to

Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard", Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., "Pulp and Paper Industries Technology and R6 Instrumentation", Kotharis Desk book series, 1995. G. Stephanopoulous, 'Chemical Process Control - Theory and Practice', R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: Sl.No Website Link 1

<https://www.nptel.ac.in/courses/103103037/>

<https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2-notes-ics-pdf-notes/> is an official topic in Module 4 of Process Control And. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed- back, controllers - proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., "Introduction to Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard", Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., "Pulp and Paper Industries Technology and R6 Instrumentation", Kotharis Desk book series, 1995. G. Stephanopoulous, 'Chemical Process Control - Theory and Practice', R7 Prentice Hall of India Ltd. R8

Lecture notes IPIR, Bangalore Online Resources: Sl.No Website Link 1

<https://www.nptel.ac.in/courses/103103037/>

<https://www.smartworld.com/notes/instrumentation-and-control-systems-pdf-2-notes-ics-pdf-notes/> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline instrumented safety system. 4 understanding the control system block diagram, negative feed-back and positive feed-back, controllers - proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. outline computerized control in process plants - programmable logic controllers - supervisory control and data acquisition (scada) - rtu (remote terminal unit) - distributed control system (dcs). final control elements - mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, i to p converter. illustrate valve safe positions, fail open and fail close. outline fail steady position and variable frequency drive [vfd] instrumented safety system- emergency shutdown system - safety interlocks. text / reference: t/r book title/author t1 eckman, d.p., industrial instrumentation, wiley eastern ltd., new york, 1990. r1 patranabis, principles of industrial instrumentation, tata mcgraw hill, 2008. ghosh a.k., "introduction to instrumentation and control", 4th ed., prentice r2 hall of india. casey j.p., "pulp and paper chemistry and chemical technology", 3rd ed., r3 vol 3, john wiley & sons. r4 robert j' measurement and control in paper making' mc grill mark r. e., "handbook of physical and mechanical testing of paper and r5 paperboard", vol. 1, marcel dekker. sankaranarayanan, p.e., "pulp and paper industries technology and r6 instrumentation", kotharis desk book series, 1995. g. stephanopoulous, 'chemical process control - theory and practice', r7 prentice hall of india ltd. r8 lecture notes ipir, banglore online resources: sl.no website link 1 <https://www.nptel.ac.in/courses/103103037/> <https://www.smartworld.com/notes/instrumentation-and-control-systems-pdf-2-notes-ics-pdf-notes/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed-back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P

Aspect	What to prepare
	converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., "Introduction to Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard", Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., "Pulp and Paper Industries Technology and R6 Instrumentation", Kotharis Desk book series, 1995. G. Stephanopoulos, 'Chemical Process Control - Theory and Practice', R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: Sl.No Website Link 1 https://www.nptel.ac.in/courses/103103037/ https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2 notes-ics-pdf-notes/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed- back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., "Introduction to Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard", Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., "Pulp and Paper Industries Technology and R6 Instrumentation", Kotharis Desk book series, 1995. G. Stephanopoulos, 'Chemical Process Control - Theory and Practice', R7 Prentice Hall of India Ltd. R8

Lecture notes IPIR, Bangalore Online Resources: Sl.No Website Link 1
<https://www.nptel.ac.in/courses/103103037/>
<https://www.smartworld.com/notes/instrumentation-and-control-systems-pdf-2>
 notes-ics-pdf-notes/ ಉಪಕರಣಗಳ ವ್ಯಾಖ್ಯಾನ/ಮಾನ್ಯತೆ ಉಪಕರಣಗಳ ವ್ಯಾಖ್ಯಾನ.
 ಉಪಕರಣಗಳ ವ್ಯಾಖ್ಯಾನ, steps, application ಉಪಕರಣಗಳ ವ್ಯಾಖ್ಯಾನ ಉಪಕರಣಗಳ. Technical
 terms English-ಉಪಕರಣಗಳ ವ್ಯಾಖ್ಯಾನ.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
 Educational explanations below are separate elaboration.

Explain the modes of control action and the computer control system
 Comprehend the principles of automatic process

M4.01		3
Understanding	control.	

M4.02	Comprehend computerized control system.	3
Understanding		

M4.03	Summarize final control elements.	4
Understanding		

M4.04	Outline instrumented safety system.	4
Understanding		

	Series Test - II	1
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Contents:

The control system block diagram, negative feed-back and positive feed- back, controllers

-proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers.

Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS).

Final control elements - Mechanism and working principle of pneumatic control

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 1 Understanding industries. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding industries*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Construct the instrumentation block diagram. (3 marks)

Answer: Begin with the standard definition or identity of *Construct the instrumentation block diagram*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic. (3 marks)

Answer: Begin with the standard definition or identity of *Suggest the functions of measuring instruments 6 Understanding Analyze different static and dynamic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 7 Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis,

resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the working principle of measuring instruments in process. (3 marks)

Answer: Begin with the standard definition or identity of 7 *Understanding characteristics of instruments Importance of instrumentation in process industries; Block diagram; Functions of measuring instruments, signaling, transmitting, registering, indicating and recording - elements of measuring instruments, primary, secondary, manipulating and functioning element. Transducers, their classification and selection criteria; On and offline measurements. Static characteristics- Accuracy, precision, sensitivity, reproducibility, drift, threshold, hysteresis, resolution, repeatability, stability, linearity, span and range, live zero, static error and dead zone Dynamic characteristics, speed of response, fidelity, lag, dynamic error. Describe the working principle of measuring instruments in process. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain 3 Understanding measuring instruments Discuss the working principle of Pressure. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding measuring instruments Discuss the working principle of Pressure. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 3 Understanding measuring instruments Outline the working principle of flow and level. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding measuring instruments Outline the working principle of flow and level. State the governing*

principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Understanding measuring instruments Illustrate the working principle of PH, viscosity. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding measuring instruments Illustrate the working principle of PH, viscosity*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Understanding and humidity measuring instruments
Temperature measuring instruments - Principle and working of -
Thermometers, filled system and vapor pressure thermometers -
Thermocouples, types, materials -resistance thermometers, thermistor -
pyrometer, radiation and optical. Pressure measuring instruments - Principle
and working of - Bourdon tube, bellows, diaphragm gauge, resistance
transducer and piezo electric pressure sensors. Flow measuring devices:
Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement
flow meters - differential pressure transducer used in venturi and orifice
meters. Level measuring devices: Float level indicator, Magnetic level
indicator, radar tank gauging system, guided wave radar tank gauging,
differential pressure cell. PH measurement, pH scale, reference and
measuring electrode - Conductivity measurement, electrical conductivity
meter - Density measurement, online density meter, hydrometer. Humidity
measurement, dew point method, wet bulb method, hygrometer, moisture-
infrared type and capacitance type Viscosity - Absolute and kinematic
viscosity - Viscosity measurement - Say bolt and Redwood viscometer -
Falling- ball viscometer.. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding and humidity measuring instruments Temperature measuring instruments - Principle and working of - Thermometers, filled system and vapor pressure thermometers -Thermocouples, types,*

materials -resistance thermometers, thermistor - pyrometer, radiation and optical. Pressure measuring instruments - Principle and working of - Bourdon tube, bellows, diaphragm gauge, resistance transducer and piezo electric pressure sensors. Flow measuring devices: Coriolis mass flow meter, Ultra sonic flow meter, and positive displacement flow meters - differential pressure transducer used in venturi and orifice meters. Level measuring devices: Float level indicator, Magnetic level indicator, radar tank gauging system, guided wave radar tank gauging, differential pressure cell. PH measurement, pH scale, reference and measuring electrode - Conductivity measurement, electrical conductivity meter - Density measurement, online density meter, hydrometer. Humidity measurement, dew point method, wet bulb method, hygrometer, moisture- infrared type and capacitance type Viscosity - Absolute and kinematic viscosity - Viscosity measurement - Say bolt and Redwood viscometer - Falling- ball viscometer.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഏകദേശം definition, പ്രവർത്തനം principle/steps, പ്രയോഗം application എന്നിവ ഉൾപ്പെടെ വിവരിക്കുക.

Module 3

Define or explain 5 Understanding paper industries. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding paper industries*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഏകദേശം definition, പ്രവർത്തനം principle/steps, പ്രയോഗം application എന്നിവ ഉൾപ്പെടെ വിവരിക്കുക.

Define or explain Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss the properties of paper 5 Understanding Illustrate the working principle for measurement*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 5 Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper. (3 marks)

Answer: Begin with the standard definition or identity of 5 *Understanding of properties of paper Definitions and methods of determination of : Physical properties - Grammage, caliper, bulk, smoothness and porosity; Mechanical - tensile strength, tear strength, burst strength, folding endurance, and bending stiffness: Optical Properties: brightness, opacity, gloss and color: Resistance Properties: water absorptiveness, Cobb test, oil absorbency, air/gas permeability. Electrical properties: Dielectric constant, Dielectric strength, Dielectric loss and Properties of electrical insulating paper.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 4

Define or explain 3 Understanding control.. (3 marks)

Answer: Begin with the standard definition or identity of 3 *Understanding control..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Comprehend computerized control system.. (3 marks)

Answer: Begin with the standard definition or identity of *Comprehend computerized control system..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Summarize final control elements.. (3 marks)

Answer: Begin with the standard definition or identity of *Summarize final control elements..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Outline instrumented safety system. 4 Understanding The control system block diagram, negative feed-back and positive feed- back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., "Introduction to Instrumentation and Control", 4th Ed., Prentice R2 Hall of India. Casey J.P., "Pulp and Paper Chemistry and Chemical Technology", 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J' Measurement and control in paper making' Mc Grill Mark R. E., "Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard",

Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., “Pulp and Paper Industries Technology and R6 Instrumentation”, Kotharis Desk book series, 1995. G. Stephanopoulous, ‘Chemical Process Control - Theory and Practice’, R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: SI.No Website Link 1 <https://www.nptel.ac.in/courses/103103037/> <https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2 notes-ics-pdf-notes/>. (3 marks)

Answer: Begin with the standard definition or identity of *Outline instrumented safety system*. 4 *Understanding The control system block diagram, negative feed-back and positive feed- back, controllers -proportional, proportional integral and proportional integral derivative controllers, on-off controllers ; analog controllers, electronic controllers and pneumatic controllers. Outline computerized Control in process plants - Programmable logic controllers - Supervisory control and data acquisition (SCADA) - RTU (Remote terminal Unit) - Distributed control system (DCS). Final control elements - Mechanism and working principle of pneumatic control valve, positioner, solenoid valve, motor operated valves, I to P converter. Illustrate valve safe positions, fail open and fail close. Outline fail steady position and variable frequency drive [VFD] Instrumented safety system- Emergency shutdown system - safety interlocks. Text / Reference: T/R Book Title/Author T1 Eckman, D.P., Industrial Instrumentation, Wiley Eastern Ltd., New York, 1990. R1 Patranabis, Principles of industrial instrumentation, Tata Mcgraw Hill, 2008. Ghosh A.K., “Introduction to Instrumentation and Control”, 4th Ed., Prentice R2 Hall of India. Casey J.P., “Pulp and Paper Chemistry and Chemical Technology”, 3rd Ed., R3 Vol 3, John Wiley & Sons. R4 Robert J’ Measurement and control in paper making’ Mc Grill Mark R. E., “Handbook of Physical and Mechanical Testing of Paper and R5 Paperboard”, Vol. 1, Marcel Dekker. Sankaranarayanan, P.E., “Pulp and Paper Industries Technology and R6 Instrumentation”, Kotharis Desk book series, 1995. G. Stephanopoulous, ‘Chemical Process Control - Theory and Practice’, R7 Prentice Hall of India Ltd. R8 Lecture notes IPIR, Bangalore Online Resources: SI.No Website Link 1 <https://www.nptel.ac.in/courses/103103037/> <https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-2 notes-ics-pdf-notes/>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□.

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Understanding industries**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding measuring instruments**
Discuss the working principle of Pressure.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **5 Understanding paper industries**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **3 Understanding control..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.nptel.ac.in/courses/103103037/>
- <https://www.smartzworld.com/notes/instrumentation-and-control-systems-pdf-notes-ics-pdf-notes/>

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